

Number Theory — Revision Summary

Divisibility, congruences, Diophantine equations, arithmetic functions, and structure

1 Divisibility, GCD, and Primes

Divisibility. For integers a, b with $a \neq 0$, $a \mid b$ means $b = ak$ for some $k \in \mathbb{Z}$. Basic properties:

- **Linearity:** $a \mid b$ and $a \mid c \Rightarrow a \mid (xb + yc)$ for all $x, y \in \mathbb{Z}$.
- **Transitivity:** $a \mid b$ and $b \mid c \Rightarrow a \mid c$.
- **Bound:** $a \mid b$ and $b \neq 0 \Rightarrow |a| \leq |b|$.

Theorem (Euclid's Lemma). If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally, if $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Theorem (Fundamental Theorem of Arithmetic). Every integer $n > 1$ has a unique factorisation $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$, up to the order of factors.

1.1 GCD, LCM, and Bézout

Key relation. $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$.

Theorem (Bézout's Identity). There exist $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$. The set $\{ax + by : x, y \in \mathbb{Z}\}$ is exactly the set of multiples of $\gcd(a, b)$.

Euclidean algorithm. For $b \neq 0$, $\gcd(a, b) = \gcd(b, a \bmod b)$. Iterate until the remainder is 0. The extended version computes the Bézout coefficients alongside.

1.2 Prime Distribution

Theorem (Dirichlet's Theorem on Arithmetic Progressions). If $\gcd(a, d) = 1$, the arithmetic progression $a, a + d, a + 2d, \dots$ contains infinitely many primes.

Theorem (Bertrand's Postulate). For every integer $n > 1$, there is always at least one prime p such that $n < p < 2n$.

Prime Number Theorem (PNT). The number of primes less than or equal to x , denoted $\pi(x)$, satisfies $\pi(x) \sim \frac{x}{\ln x}$ as $x \rightarrow \infty$.

2 Modular Arithmetic and Congruences

Congruence. $a \equiv b \pmod{m}$ means $m \mid (a - b)$. Congruence respects $+$, $-$, $\times$:

$$a \equiv b, c \equiv d \pmod{m} \Rightarrow a \pm c \equiv b \pm d, ac \equiv bd \pmod{m}.$$

- **Cancellation:** If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.
- **Linear congruence:** $ax \equiv b \pmod{m}$ has a solution iff $\gcd(a, m) \mid b$. If $d = \gcd(a, m)$, there are exactly d incongruent solutions modulo m .
- **Inverses:** a has an inverse mod m iff $\gcd(a, m) = 1$.

2.1 Euler Totient, Fermat, Wilson

Euler's totient $\varphi(n)$. The count of integers in $[1, n]$ coprime to n . If $n = \prod p_i^{\alpha_i}$:

$$\varphi(n) = n \prod_i \left(1 - \frac{1}{p_i}\right).$$

Multiplicative: $\gcd(m, n) = 1 \Rightarrow \varphi(mn) = \varphi(m)\varphi(n)$.

Theorem (Fermat's Little Theorem). For prime p and $p \nmid a$: $a^{p-1} \equiv 1 \pmod{p}$. Equivalently $a^p \equiv a \pmod{p}$ for every integer a .

Theorem (Euler's Theorem). If $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$. In particular $a^{-1} \equiv a^{\varphi(n)-1} \pmod{n}$.

Theorem (Wilson's Theorem). p is prime iff $(p-1)! \equiv -1 \pmod{p}$.

2.2 Multiplicative Order and Primitive Roots

Order $\text{ord}_n(a)$. The least $k \geq 1$ with $a^k \equiv 1 \pmod{n}$ (defined when $\gcd(a, n) = 1$).

- $a^m \equiv 1 \pmod{n}$ iff $\text{ord}_n(a) \mid m$.
- $\text{ord}_n(a) \mid \varphi(n)$.
- $\text{ord}_n(a^t) = \frac{\text{ord}_n(a)}{\gcd(\text{ord}_n(a), t)}$.

Primitive root. g is a primitive root mod n if $\text{ord}_n(g) = \varphi(n)$. Such a g exists iff $n \in \{1, 2, 4, p^k, 2p^k\}$ for p an odd prime, $k \geq 1$. When a primitive root exists, the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is cyclic of order $\varphi(n)$.

2.3 Polynomial Congruences and Hensel's Lemma

Lemma (Thue's Lemma). For any integer a and modulo m , there exist integers x, y such that $0 < |x| \leq \sqrt{m}$, $0 < |y| \leq \sqrt{m}$, and $x \equiv ay \pmod{m}$.

Theorem (Hensel's Lemma). Let $f(x)$ be a polynomial with integer coefficients. If $a \in \mathbb{Z}$ such that $f(a) \equiv 0 \pmod{p^k}$ and $f'(a) \not\equiv 0 \pmod{p}$, then there exists a unique integer $b \pmod{p^{k+1}}$ such that $f(b) \equiv 0 \pmod{p^{k+1}}$ and $b \equiv a \pmod{p^k}$.

Use. The primary tool for lifting solutions from modulo p to modulo p^k .

3 Chinese Remainder Theorem

Theorem (CRT, pairwise coprime moduli). If $m_1, \dots, m_r$ are pairwise coprime, the system $x \equiv a_i \pmod{m_i}$ ($1 \leq i \leq r$) has a unique solution modulo $M = m_1 \cdots m_r$.

Constructive form. Let $M_i = M/m_i$ and $y_i \equiv M_i^{-1} \pmod{m_i}$. Then

$$x \equiv \sum_{i=1}^r a_i M_i y_i \pmod{M}.$$

Use. CRT splits a problem mod M into independent sub-problems mod each prime power. Almost every olympiad problem about “find all n such that...” with composite modulus is attacked this way.

4 p -adic Valuations and LTE

Valuation ν_p . For prime p and $n \neq 0$, $\nu_p(n)$ is the exponent of p in the prime factorisation of n .

- $\nu_p(ab) = \nu_p(a) + \nu_p(b)$.
- $\nu_p(a^k) = k\nu_p(a)$ for $k \geq 0$.
- $\nu_p(a+b) \geq \min\{\nu_p(a), \nu_p(b)\}$, with equality whenever $\nu_p(a) \neq \nu_p(b)$.

Legendre's formula. $\nu_p(n!) = \sum_{k \geq 1} \lfloor n/p^k \rfloor = \frac{n - s_p(n)}{p-1}$, where $s_p(n)$ is the digit sum of n in base p .

4.1 Lifting the Exponent (LTE)

A precise valuation lemma for $\nu_p(x^n \pm y^n)$.

Theorem (LTE, odd prime, minus form). If p is an odd prime, $p \mid (x - y)$, $p \nmid xy$, then for $n \geq 1$:

$$\nu_p(x^n - y^n) = \nu_p(x - y) + \nu_p(n).$$

Theorem (LTE, odd prime, plus form). If p is an odd prime, $p \mid (x + y)$, $p \nmid xy$, and n is odd:

$$\nu_p(x^n + y^n) = \nu_p(x + y) + \nu_p(n).$$

Theorem (LTE for $p = 2$). If x, y are both odd and n is even:

$$\nu_2(x^n - y^n) = \nu_2(x - y) + \nu_2(x + y) + \nu_2(n) - 1.$$

Use. The standard tool whenever a problem asks for the exact power of p dividing $a^n - b^n$ or $a^n + b^n$. Common applications: existence/non-existence of solutions to $a^n + b^n = c^k$, periodicity of orders mod p^k .

5 Arithmetic Functions

Multiplicative functions. A function $f : \mathbb{N} \rightarrow \mathbb{C}$ is multiplicative if $f(mn) = f(m)f(n)$ for all coprime m, n . It is completely multiplicative if this holds for all m, n . Common functions: $\varphi(n)$ (totient), $d(n)$ (number of divisors), $\sigma(n)$ (sum of divisors), $\mu(n)$ (Möbius).

Sum-of-divisors and number-of-divisors. For $n = \prod p_i^{\alpha_i}$:

$$d(n) = \prod (\alpha_i + 1), \quad \sigma(n) = \prod \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

Both are multiplicative.

5.1 Dirichlet Convolution and Möbius Inversion

Dirichlet Convolution. Given two arithmetic functions f and g , their Dirichlet convolution is:

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

The identity element is $I(n) = \lfloor 1/n \rfloor$.

Möbius function. $\mu(n) = (-1)^k$ if n is a product of k distinct primes, 0 if n has a squared prime factor, $\mu(1) = 1$. Key property: $\sum_{d|n} \mu(d) = I(n)$.

Theorem (Möbius Inversion Formula). For arithmetic functions f and g : $f(n) = \sum_{d|n} g(d) \iff g(n) = \sum_{d|n} \mu(n/d)f(d)$.

In terms of Dirichlet convolution, $f = g * 1 \iff g = f * \mu$.

6 Diophantine Equations

6.1 Linear Diophantine Equations

Theorem (Solvability). $ax + by = c$ has integer solutions iff $\gcd(a, b) \mid c$. If $d = \gcd(a, b)$ and (x_0, y_0) is one solution, all solutions are

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t, \quad t \in \mathbb{Z}.$$

Find (x_0, y_0) via the extended Euclidean algorithm to get Bézout coefficients for d , then multiply by c/d .

6.2 Non-Linear Diophantine Equations

Pythagorean Triples. Solutions to $x^2 + y^2 = z^2$. A triple is primitive if $\gcd(x, y, z) = 1$. All primitive Pythagorean triples (with x even) are given by:

$$x = 2mn, \quad y = m^2 - n^2, \quad z = m^2 + n^2$$

where $m > n > 0$, $\gcd(m, n) = 1$, and exactly one of m, n is even.

Pell's Equation. $x^2 - dy^2 = 1$, where d is a positive non-square integer. Always has infinitely many integer solutions. Let (x_1, y_1) be the fundamental solution (the solution with the smallest positive x and y). All positive solutions (x_k, y_k) are generated by:

$$x_k + y_k\sqrt{d} = (x_1 + y_1\sqrt{d})^k, \quad k = 1, 2, 3, \dots$$

The negative Pell's equation $x^2 - dy^2 = -1$ has solutions only for some d . If it has a fundamental solution (u_1, v_1) , all its solutions are odd powers of $u_1 + v_1\sqrt{d}$.

Pell Matrix. The recurrence for solutions (x_k, y_k) can be expressed elegantly using matrix multiplication. Given the fundamental solution (x_1, y_1) to $x^2 - dy^2 = 1$, subsequent solutions satisfy:

$$\begin{pmatrix} x_k \\ y_k \end{pmatrix} = \begin{pmatrix} x_1 & dy_1 \\ y_1 & x_1 \end{pmatrix}^k \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 & dy_1 \\ y_1 & x_1 \end{pmatrix} \begin{pmatrix} x_{k-1} \\ y_{k-1} \end{pmatrix}.$$

This provides a fast method to compute large solutions or reduce them modulo a prime via matrix exponentiation.

Method of Infinite Descent. A proof technique to show an equation has no solutions (or only trivial ones). Assume a positive integer solution exists, then construct a strictly smaller positive integer solution, leading to a contradiction. *Fermat's Last Theorem for $n = 4$:* $x^4 + y^4 = z^2$ has no non-trivial integer solutions, proved by infinite descent.

Vieta Jumping. A specific type of infinite descent used on relations of the form $x^2 - kxy + y^2 = m$. If (x, y) is a solution, consider it as a quadratic in x . By Vieta's formulas, the other root x' gives a new solution $(x', y) = (ky - x, y)$. Analysing the size of x' often forces a contradiction or bounds the solutions.

7 Quadratic Residues

QR and Legendre symbol. a is a quadratic residue mod m if $x^2 \equiv a \pmod{m}$ has a solution. For odd prime p , the Legendre symbol is

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & a \text{ is a nonzero QR mod } p, \\ -1 & a \text{ is a non-QR mod } p, \\ 0 & p \mid a. \end{cases}$$

Theorem (Euler's Criterion). $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.

Theorem (Quadratic Reciprocity). For distinct odd primes p, q :

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Supplementary laws: $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$.

Jacobi Symbol. A generalisation of the Legendre symbol to odd composite numbers $n = \prod p_i^{\alpha_i}$:

$$\left(\frac{a}{n}\right) = \prod \left(\frac{a}{p_i}\right)^{\alpha_i}$$

Follows the same reciprocity laws as the Legendre symbol, allowing rapid calculation without factoring n . However, $\left(\frac{a}{n}\right) = 1$ does *not* guarantee a is a QR mod n .

Use. The supplementary laws plus reciprocity reduce any Legendre/Jacobi symbol to small cases via repeated application. Whenever a problem asks “is the equation $x^2 \equiv a \pmod{p}$ solvable?” the answer is computable in seconds.

8 Continued Fractions and Farey Sequences

Farey Sequence $\mathcal{F}_n$. The sequence of completely reduced fractions between 0 and 1 with denominators $\leq n$, ordered by size. Properties:

- If a/b and c/d are adjacent in $\mathcal{F}_n$, then $bc - ad = 1$.
- The mediant $\frac{a+c}{b+d}$ is the unique fraction with the smallest denominator strictly between a/b and c/d .

Continued Fractions. Any real number α can be expressed as:

$$\alpha = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}} = [a_0; a_1, a_2, \dots]$$

The truncations $[a_0; a_1, \dots, a_n] = p_n/q_n$ are the *convergents* of α .

- Convergents are the best rational approximations to α .
- The continued fraction of α is finite iff $\alpha \in \mathbb{Q}$.
- The continued fraction of α is eventually periodic iff α is a quadratic irrational.
- *Pell Connection:* The fundamental solution to $x^2 - dy^2 = 1$ is always found among the convergents of $\sqrt{d}$.

9 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
<i>Divisibility and Structure</i>	
gcd, Bézout coefficients, modular inverse	Extended Euclidean algorithm
System $x \equiv a_i \pmod{m_i}$, coprime m_i	Chinese Remainder Theorem (CRT)
Composite modulus problem	Split into prime-power moduli, then CRT
Highest power of p dividing $n!$	Legendre's formula
Highest power of p dividing $a^n \pm b^n$	Lifting the Exponent Lemma (LTE)
Lift solution $f(x) \equiv 0 \pmod{p}$ to $\pmod{p^k}$	Hensel's Lemma
Primes in arithmetic progression $an + b$	Dirichlet's Theorem (if $\gcd(a, b) = 1$)
Multiplicative function summed over divisors	Dirichlet Convolution, Möbius Inversion
Counting cyclic subgroup elements mod p	Primitive root
Order of a mod n	Divides $\varphi(n)$; check divisors
<i>Congruences and Residues</i>	
Reduce $a^k \pmod{p}$ for prime p	Fermat's Little Theorem: $a^{p-1} \equiv 1$
Reduce $a^k \pmod{n}$, $\gcd(a, n) = 1$	Euler's Theorem: $a^{\varphi(n)} \equiv 1$
Is $(p-1)! \equiv -1 \pmod{p}$?	Wilson's Theorem (iff p is prime)
Is $x^2 \equiv a \pmod{p}$ solvable?	Legendre symbol + Quadratic Reciprocity
Fast calculation of $\left(\frac{a}{p}\right)$ without factoring	Jacobi symbol generalisation
$x^2 \equiv a \pmod{p}$ given $(a, p) = 1$	Solutions exist iff $a^{(p-1)/2} \equiv 1 \pmod{p}$
<i>Diophantine Equations</i>	
Linear: $ax + by = c$	Solvable iff $\gcd(a, b) \mid c$
Quadratic sum: $x^2 + y^2 = z^2$	Primitive Pythagorean triples parametrisation
$x^2 - dy^2 = 1$ (d non-square)	Pell's Equation; Pell Matrix / convergents
$x^2 - kxy + y^2 = m$	Vieta Jumping (quadratic in x)
Prove no positive integer solutions exist	Method of Infinite Descent
Best rational approximation of real α	Continued fraction convergents
Fractions $\frac{a}{b}$ in $[0, 1]$ with $b \leq n$	Farey Sequence properties ($bc - ad = 1$)

Three universal moves for Number Theory: (1) *Factor and lift.* Reduce mod prime powers (CRT), apply Fermat / Euler / LTE / Hensel, and recombine. (2) *Bounding and inequalities.* Use the absolute value bound ($a \mid b \implies |a| \leq |b|$) or asymptotic growth to severely restrict possible integer solutions. (3) *Symmetry in quadratics.* For degree-two Diophantine equations, treat the equation as a quadratic in one variable and use the discriminant or Vieta's formulas to generate new solutions or bound existing ones.